

6. Applications of Integrals

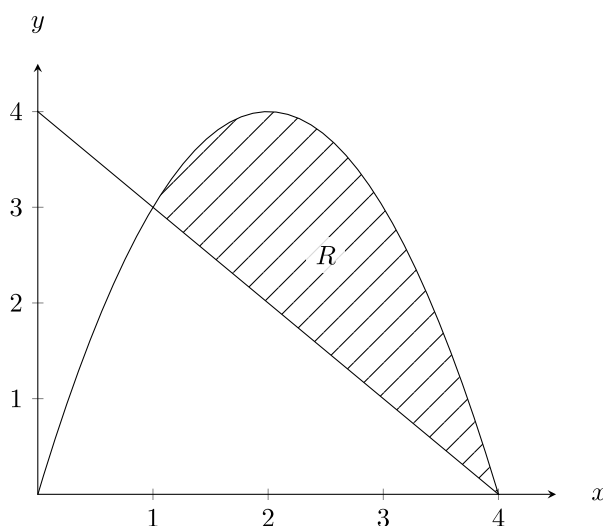
6.1. Area Between Curves

6.1 Learning Objectives

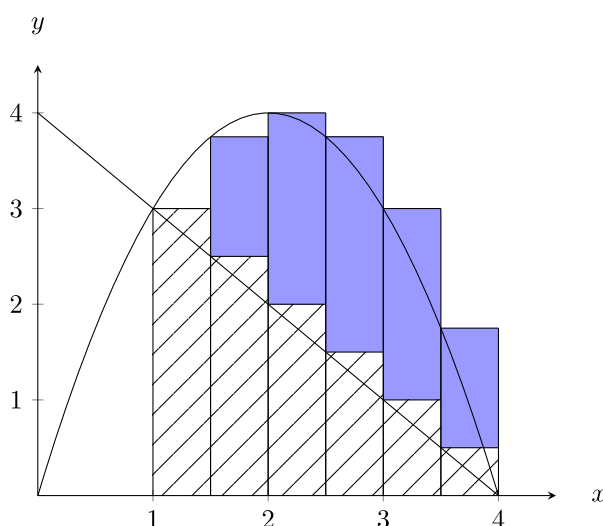
- I can extend the idea of area under a curve to area between curves.

Suppose we wish to find the area between the curves $y = -x^2 + 4x$ and $y = -x + 4$ on $[1, 4]$.

How can we approximate this area?



To approximate this area, we can draw rectangles between the curves, with bottom edge lined up with the lower curve, and top edge lined up with the upper curve. This is effectively the same as estimating the area under each curve, and taking the difference between them (with the same number of rectangles). Note in the following plot, the first blue rectangle is the same height as the dashed one (because the functions intersect at the left side of this sub-interval), so it is not displayed.



We can approximate the area under both curves individually. Then, since the area under the “lower” curve is under both curves, we can subtract it away to get the area that is left.

Definition 6.1.1 Area Between Curves

Let R denote the region between continuous functions $y = f(x)$ and $y = g(x)$ and between the vertical lines $x = a$ and $x = b$, given that $f(x) \geq g(x)$ on $[a, b]$. The area of R is defined by

$$\begin{aligned} A &= \left(\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x \right) - \left(\lim_{n \rightarrow \infty} \sum_{i=1}^n g(x_i^*) \Delta x \right) \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n (f(x_i^*) - g(x_i^*)) \Delta x. \end{aligned}$$

where we divide R into n strips of equal width, $\Delta x = \frac{b-a}{n}$ and height $f(x_i^*) - g(x_i^*)$. Thus, written as a definite integral, $A = \int_a^b (f(x) - g(x)) \, dx$.

Note

It isn't really necessary to define the area between curves in terms of Reimann sums. So long as you can reason out why the area between curves is the integral of their difference, you are good to go. You do also need to know the limit of Reimann sums definition of the integral, so it is good practice to go over it again here in a slightly more complicated format. Discuss in office hours if you have questions about this formulation.

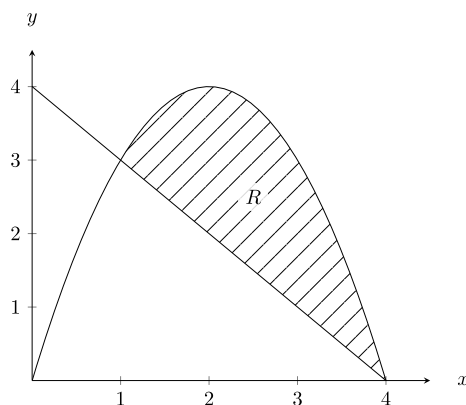
Exercise 6.1.1

Find the area between the curves $y = -x^2 + 4x$ and $y = -x + 4$ on $[1, 4]$.

Solution

- It is good to sketch the region in these cases, and find any intersection points. Remember that we only get the area between curves if the upper function is, well, the upper function (we require $f(x) \geq g(x)$ on $[a, b]$ in the area between curves definition). If the functions changed which one was greater than the other over our interval, we would just break up the interval and compute a few integrals.

The region R was graphed before but is reproduced here for completion:



Note, on $[1, 4]$ we always have $-x^2 + 4x \geq -x + 4$. Then, the area, A , between curves is

$$\begin{aligned}
 A &= \int_1^4 (-x^2 + 4x) - (-x + 4) \, dx \\
 &= \int_1^4 -x^2 + 5x - 4 \, dx \\
 &= \left[-\frac{1}{3}x^3 + \frac{5}{2}x^2 - 4x \right]_1^4 \\
 &= -\frac{1}{3}(4)^3 + \frac{5}{2}(4)^2 - 4(4) - \left(-\frac{1}{3}(1)^3 + \frac{5}{2}(1)^2 - 4(1) \right) \\
 &= \frac{19}{2}
 \end{aligned}$$

Exercise 6.1.2

1. Compute the area between the curves $y = -x^2 + 4x$ and $y = -x + 4$ on $[0, 2]$. *Note: you will need to compute this using 2 definite integrals, make sure to check that the upper function is the first one in your difference.*
2. Convince yourself that the area between curves definition works even when $g(x) \leq 0$. You should be finding the *unsigned* area when you do this, what does that tell you about the sign of your answer to a problem asking for the area between curves?

Instead of taking vertical rectangles to approximate the area between curves, we might look at horizontal rectangles to find area between curves. This is especially useful in cases where we have more of a “right” and “left” curve instead of an “upper” and “lower” curve, as in the following example.

Example 6.1.1

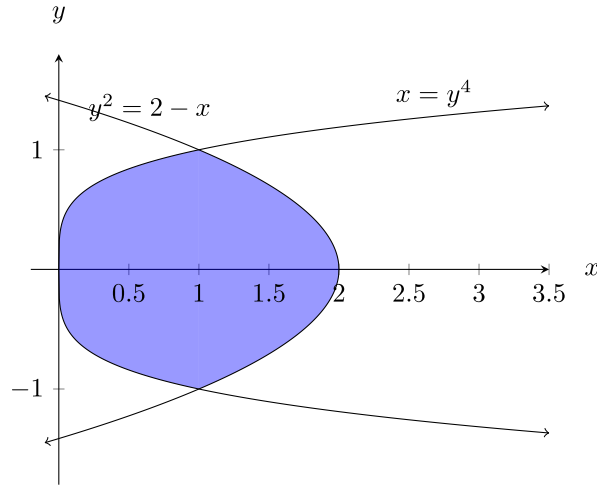
Set up and evaluate an integral to find the area enclosed by $x = y^4$ and $y^2 = 2 - x$.

Warning

It is expected that you know how to graph basic functions and similar relations like the ones above, although you may not have been formally trained in a class on these specifically. Please come discuss this in office hours if the hint to “turn your head sideways and think about swapping x and y ” does not help you. It is perfectly reasonable to be a little stuck on this when you first see it, and that is what office hours are there for!

Solution

First, we sketch the region to get an idea of what it looks like.



We see we have more of a right curve ($y^2 = 2 - x$) and left curve ($x = y^4$). In terms of x values, we have $x = y^4$ and $x = 2 - y^2$.

We start by finding where the functions intersect

$$\begin{aligned} y^4 &= 2 - y^2 \\ y^4 + y^2 - 2 &= 0 \\ (y^2 + 2)(y^2 - 1) &= 0 \end{aligned}$$

since $y^2 + 2$ is never 0, we know they intersect when $y^2 - 1 = 0$. To find the area between them, we can think about turning our head sideways and doing an integral in y , where instead of “upper function minus lower function” in the integral, we do “right function minus left function” (since $2 - y^2 \leq y^4$ for y on $[-1, 1]$ our method still applies). Thinking of this as an integral in y we have

$$\begin{aligned} \int_{-1}^1 (2 - y^2) - y^4 \, dy &= \left[2y - \frac{1}{3}y^3 - \frac{1}{5}y^5 \right]_{-1}^1 \\ &= 2 - \frac{1}{3} - \frac{1}{5} - \left(-2 + \frac{1}{3} + \frac{1}{5} \right) \\ &= \frac{44}{15} \end{aligned}$$

Example 6.1.2

Set up an integral to find the area enclosed between $f(x) = \cos(x)$ and $g(x) = \sin(x)$ on $[0, 2\pi]$.

Solution

Sketching the graphs on $[0, 2\pi]$ and using our knowledge of the unit circle, we know that the functions intersect at $x = \frac{\pi}{4}, \frac{5\pi}{4}$. For $[0, \frac{\pi}{4}]$, $f(x) \geq g(x)$ and then on $[\frac{\pi}{4}, \frac{5\pi}{4}]$, $g(x) \geq f(x)$, then at last on $[\frac{5\pi}{4}, 2\pi]$, $f(x) \geq g(x)$ once again. So, the area between the curves, A , is given by

$$A = \int_0^{\frac{\pi}{4}} (\cos(x) - \sin(x)) \, dx + \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin(x) - \cos(x)) \, dx + \int_{\frac{5\pi}{4}}^{2\pi} (\cos(x) - \sin(x)) \, dx$$

Tip

To find the area between 2 curves do the following

1. Identify the domain, so you have a good idea of what the region is.
2. Find where the curves intersect: set the equations of the curves equal to each other, and make sure to consider any domain restrictions.
3. Determine the subintervals over which you need to calculate the area: does the “upper curve” switch over the interval?
4. Determine whether $f(x) \geq g(x)$ or $g(x) \geq f(x)$ on each sub-interval.
5. Set up the integrals for each sub-interval with the greater function minus the smaller function (“upper minus lower” or “right minus left”) and evaluate.
6. Check: We are looking for *unsigned* area when we ask for the area between the curves, you should always get a positive number!

Exercise 6.1.3

Set up an integral to find the area of the triangular region enclosed by the points $(0, 0)$, $(2, 2)$, $(5, 0)$ with respect to both x and y .

6.1 Section Summary:

- We learned how to find the area between curves, either in the x or y direction.